



Sri Lanka Institute of Information Technology

B.Sc. (Hons) in IT - Field of Specialization Interactive Media

2026 - Year 04 - Semester 02

SE4051 - Trends in Digital Media

Designing an Intelligent and Human-Centered
Interactive Environment

Self-Study Explanation

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1. Eco-Feedback System in an Interactive Beach Cleanup Environment

For the self-study innovation component, I explored the concept of an eco-feedback system and how it can be used in an interactive digital media environment. Eco-feedback refers to giving users clear and immediate feedback about their actions related to environmental responsibility. Instead of only telling users that pollution is harmful, an eco-feedback system allows users to experience the result of positive or negative behavior through interaction, visual feedback, audio feedback, and progress indicators.

I selected this concept because my project is based on a Sri Lankan beach cleanup experience. Beaches are important natural and social spaces in Sri Lanka, but plastic waste and careless human behavior can damage coastal environments and marine life. Therefore, I wanted to create an interactive first-person experience where the user actively participates in cleaning the beach rather than simply reading information about environmental protection.

In my Unity project, the player explores a tropical beach environment and collects trash items scattered across the sand. When the player moves near a trash object, the system displays a clear “Press E” interaction prompt. After collecting the trash, the object disappears, a collection sound plays, and the score increases. This provides immediate feedback and rewards the player for environmentally positive action. I also added a trash bag power-up that temporarily gives a score multiplier, making the experience more engaging and game-like.

The system also includes negative feedback. If the player moves outside the safe area, a danger warning appears with a warning sound, and the player is returned to safety. This helps guide the user and keeps the experience understandable. The timer, score UI, tutorial screen, pickup prompts, warning effects, and beach sound effects all work together to create a responsive environment. These features support the assignment requirement for intelligent behavior, adaptive responses, and user-centered design.

Through this self-study, I learned that intelligent interaction does not always require complex artificial intelligence. Simple rule-based systems can still feel intelligent if they respond clearly to user behavior. I also learned that feedback is very important in digital media design. When the player receives visual and audio responses immediately after an action, the environment feels more alive and meaningful. This project helped me understand how interactive media can be used not only for entertainment, but also for awareness, education, and behavior change.



Figure 1.1 - Eco-feedback through trash collection prompt



Figure 1.2 - Adaptive danger warning when leaving the safe area